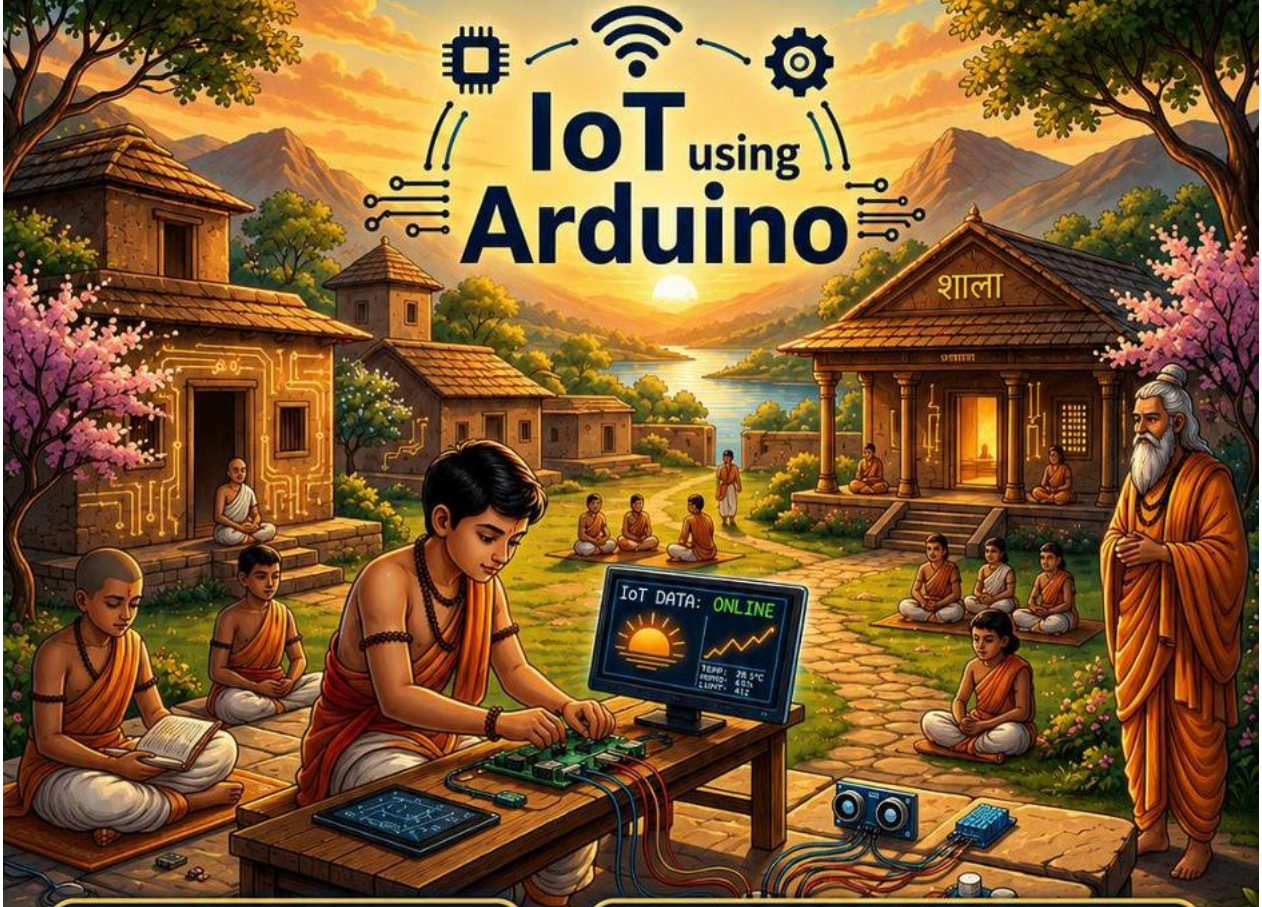
TECH OLYMPIAD 2026

विद्या भारती उत्तर क्षेत्र
VIDYA BHARATI UTTAR KSHETRA

Book-1

For Classes 6-8 & 9-10

IoT using Arduino



FOR CLASSES

6-8 & **9-10**

ABOUT THE SERIES

We are conducting exam for 2 class groups:

- Classes 6-8
- Classes 9-10

Book-1 is for both groups.
Book-2 is for Classes 9-10 only.



A QUICK HELLO

Have you ever wondered how a door opens **automatically** when you walk towards it, how a plant gets water by itself when the soil is dry, or how machines sense things and take action?

EXAMPLE



Smart Plant Watering System

Welcome to the exciting world of **IoT – Internet of Things!**



In this book, you will learn how everyday objects can become smart using sensors, electronic components and programming. A device can **sense**, **think** and **act**—just like a tiny electronic brain!

At the heart of this book is **Arduino**, a small but powerful board. Connect **sensors** (eyes and ears), and **lights**, **buzzers**, **motors** (hands and feet) and make your projects come alive!



You will learn about Arduino, sensors, LEDs, motors, switches and many more components. You will also write simple programs and see them work in real life!

With **Tinkercad**, you can build and test circuits virtually before making them for real!

Ask questions, try new ideas, make mistakes and learn.

Your curiosity will turn into creativity and innovation!

LEARN | EXPERIMENT | CREATE | INNOVATE

Open the book, start experimenting and let your imagination become technology!



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1. Understanding IoT

1.1 Introduction

Future is arriving faster than ever, transforming how children learn, play, and interact with the world. The concept of the Internet of Things (IoT) has emerged not just as a futuristic idea but as a part of our daily lives.

By integrating IoT with the innate curiosity and adaptability of kids, we're opening a doorway to a future where every young mind understands how to make everyday objects smarter with code.

Vidya Bharti Tech Olympiad 2026, explores how IoT can be both a playground and a classroom for kids, turning their interactions with technology into opportunities for learning and innovation.

1.2 What is IoT?

Have you ever thought that objects around you could talk to each other?

Just Imagine,

- if your school bag could remind you that you forgot your maths notebook, or
- your refrigerator might order more pizza before you even know it's gone?, or
- if your home knows which lights to turn off when you leave the room, or
- If your toy car sent a text to your tablet saying, " Hey! I'm sleepy! I need new batteries'? " or
- if your water bottle told you, "Hey, drink some water, you are feeling thirsty!", or
- If your school bus told your parents exactly where it is on the map?

That's not magic — it's the 'Internet of Things' (IoT).

In simple words, IoT is a network where everyday objects are embedded with technology allowing them to send, receive and process data to make decisions.

IoT systems are networks of physical devices that collect data, share information, and make decisions. Think of it as giving everyday objects a brain and a voice.

These objects use small parts called sensors to feel things, internet to talk, and apps to show us the information. So, your things don't just sit quietly — they become smart helpers for you!

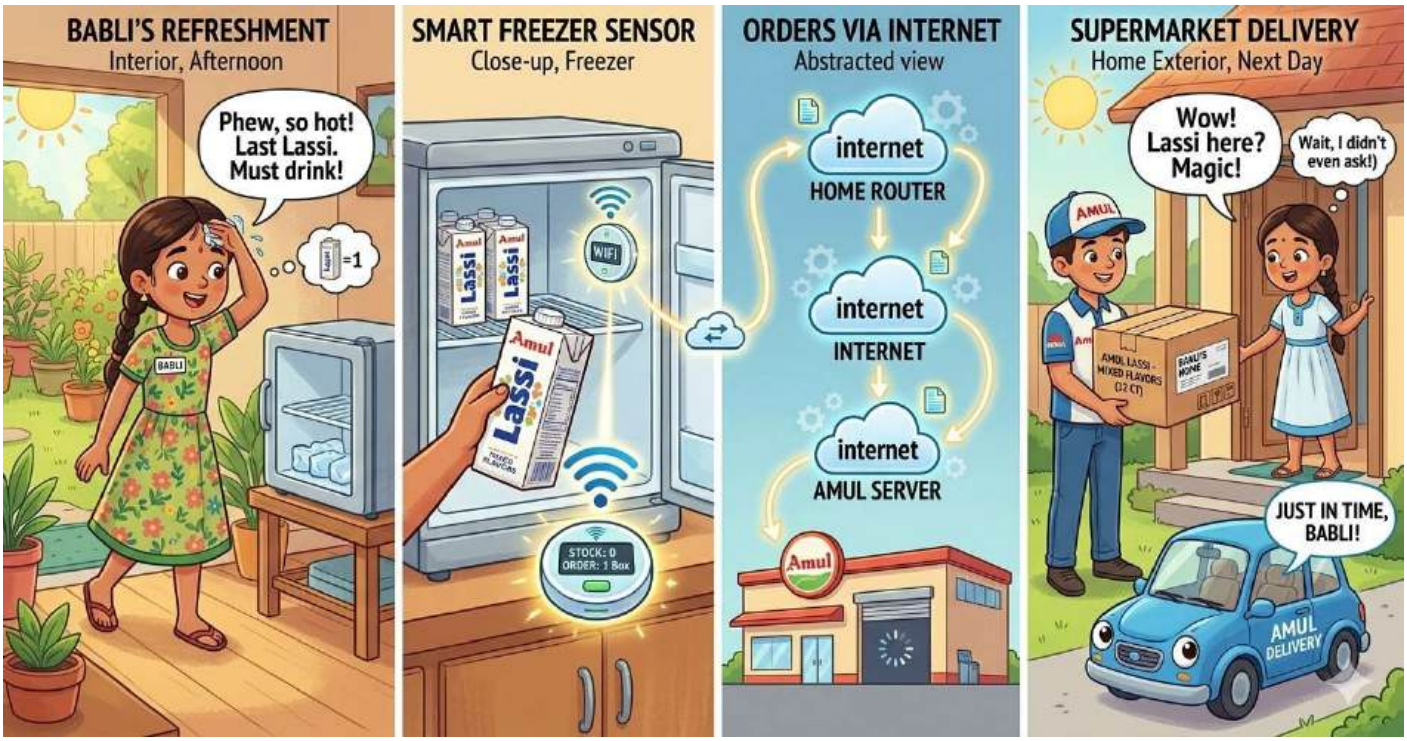


1.3 IoT Stories

Story-1

Imagine,

- Babli come in from playing in the garden and it's hot,
- she go to the freezer and take out an Amul Lassi– it's the last one.
- Instead of asking her parents to buy more, her freezer talks over the internet to the supermarket and
- the next day her new box of Amul Lassi arrives, almost like magic.

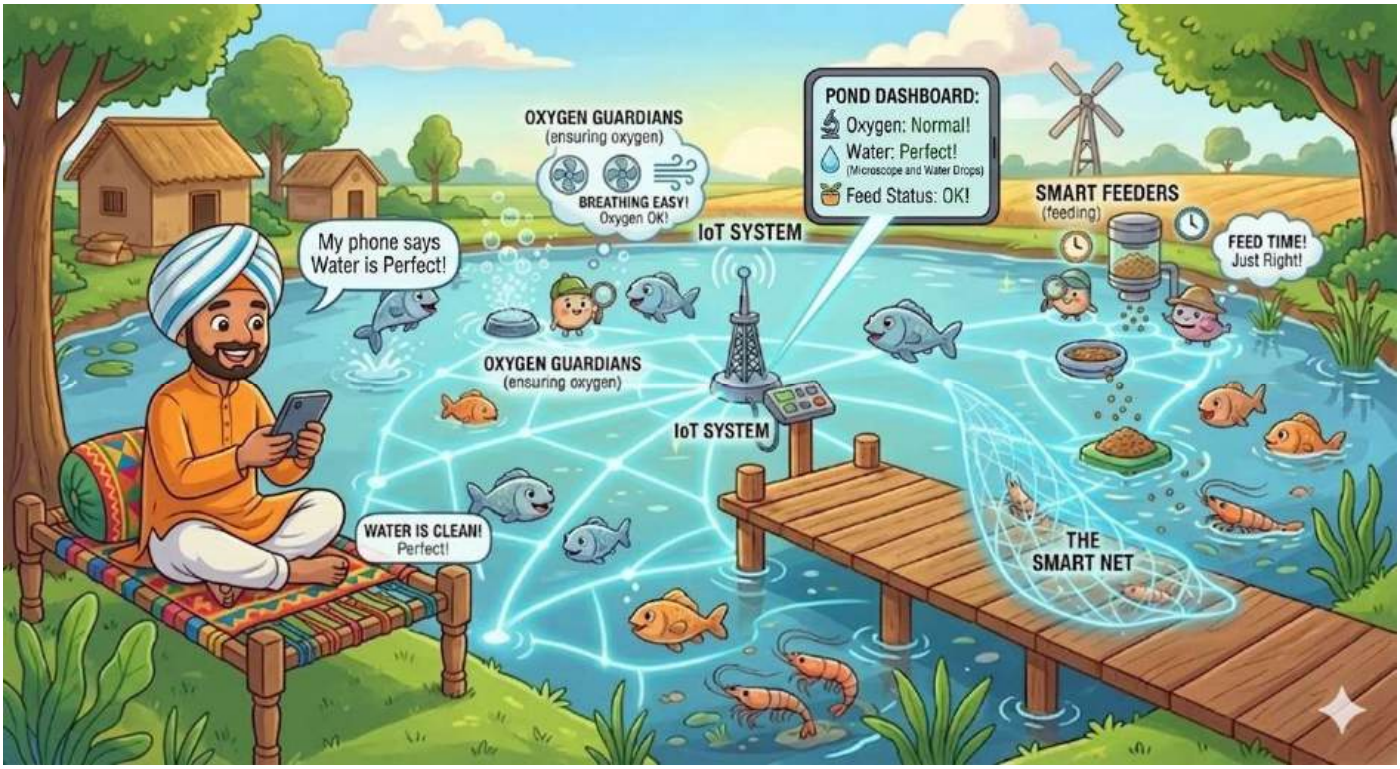




Story-2

Imagine

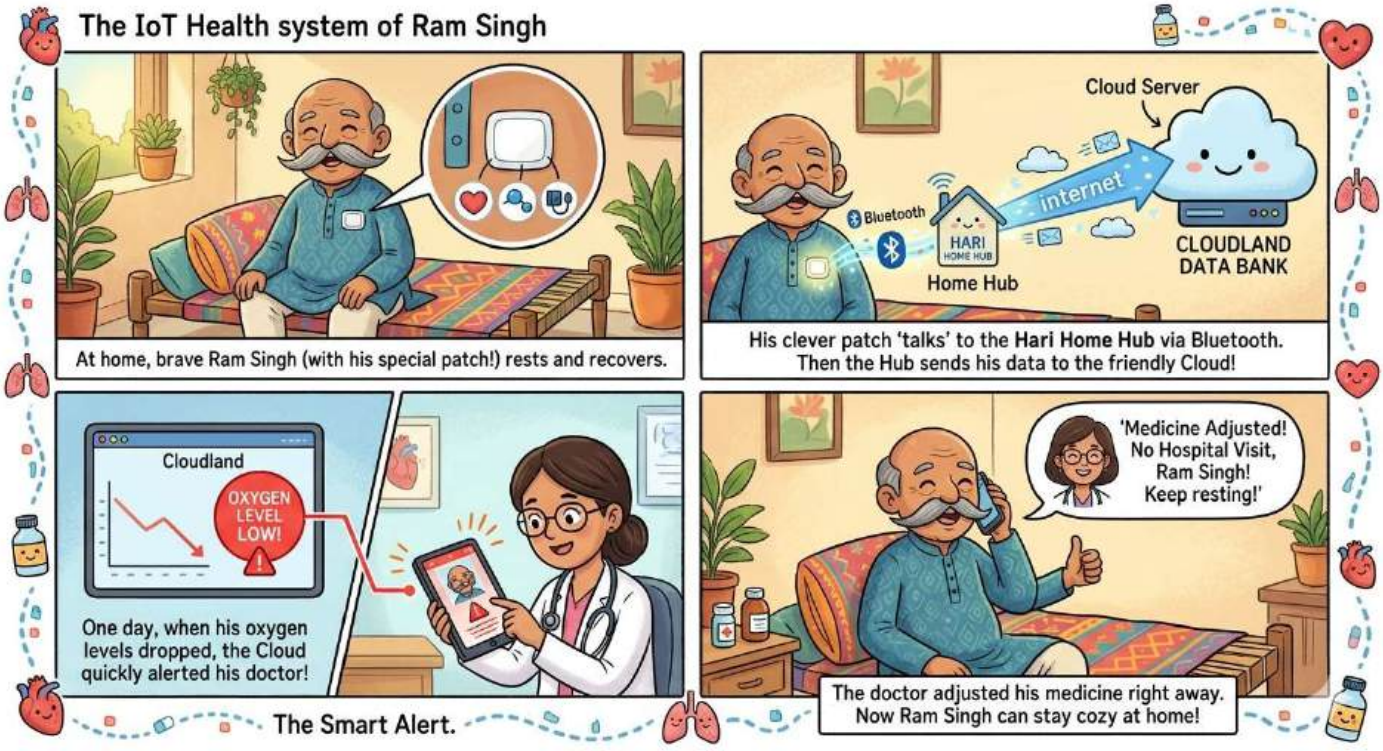
- a big, cozy farm for fish and shrimp
- but instead of walking around, the farmer uses a magical "Smart Net" (the Internet of Things - IoT) to take care of them.
- This magic net uses tiny sensors that behave like detectives to keep the
  - water perfectly clean,
  - ensure the fish have enough oxygen to breathe, and
  - even feed them exactly when they are hungry
- This system ensures the animals grow up healthy and strong.



Story-3

Imagine

- 70-year-old Ram Singh recovering at home from heart failure
- He wears a smart patch that constantly tracks his heart rate, oxygen levels, and blood pressure,
- Transmitting this data via Bluetooth to his home hub and then to the cloud.
- When his oxygen saturation dropped, an IoT-powered system alerted his cardiologist immediately,
- allowing for medication adjustments without a hospital visit.





1.4 This is Not IoT





2. Main Components of an IoT System

2.1 Sensors: Eyes & Ears

Sensors gather information from the world around them. They're like your senses—seeing, hearing, feeling temperature, or detecting movement. These are the elements responsible for collecting raw data from the environment.

A device can have one sensor or several sensors that function collectively. These sensors change the physical parameters into digital signals that IoT platforms can understand.

Common Types of Sensors you Should Know:

- Temperature Sensors, measure heat, (think smart thermostats)
- Motion Sensors, detect when something moves (like automatic doors)
- Light Sensors, measure brightness (your phone's auto-brightness uses this)
- Sound Sensors, pick up noise levels (voice assistants listen with these)

Real Example:

When you walk into a room and the lights turn on automatically, a motion sensor detected you and sent that information to the lights.





2.2 Connectivity: The Bridge That Links Devices to the Cloud

Once sensors collect data, the information must travel somewhere for processing. This is where connectivity comes in—a messenger delivering information. This is how devices talk to each other and share the data collected by sensors.

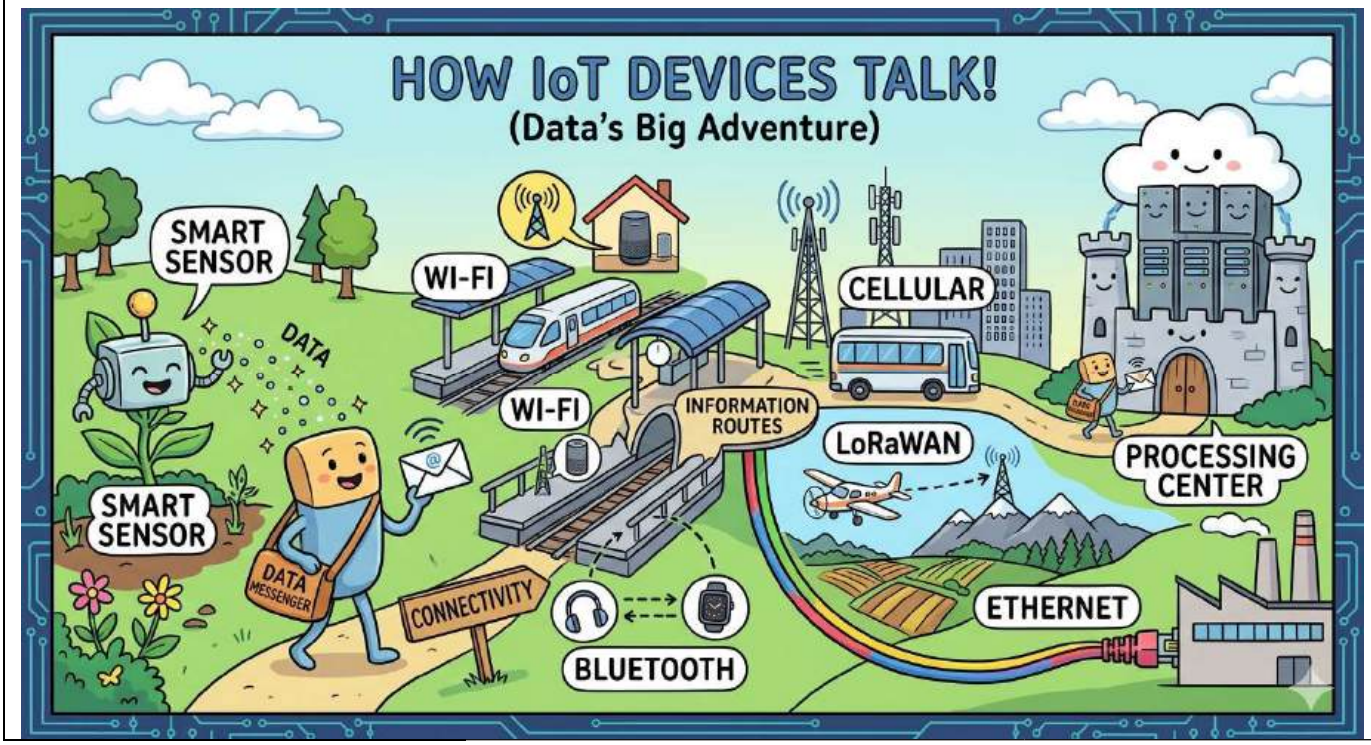
Depending on the application, IoT devices can use various communication technologies, including

- **Wi-Fi**: Ideal for indoor, high-bandwidth communication(most home devices use this)
- **Bluetooth / BLE**: Best for short-range energy-efficient transmission (wireless earbuds connect this way)
- **Cellular**: Suitable for mobile or wide-area IoT applications (smartwatches work without your phone)
- **LoRaWAN:** Designed for long-range, low-power communication
- **Ethernet** – Preferred in industrial settings requiring stable connections

Why Connectivity Matters:

Without connectivity, sensors would collect data but have nowhere to send it. It's like taking photos but never sharing them with anyone. Proper IoT connectivity ensures devices communicate reliably and securely.

Poor connectivity can cause data loss, slowdowns, and irregular monitoring—that negatively affect the whole decision-making process in IoT systems like labs, manufacturing plants, or hospitals etc.





2.3 Data Processing: The Brain

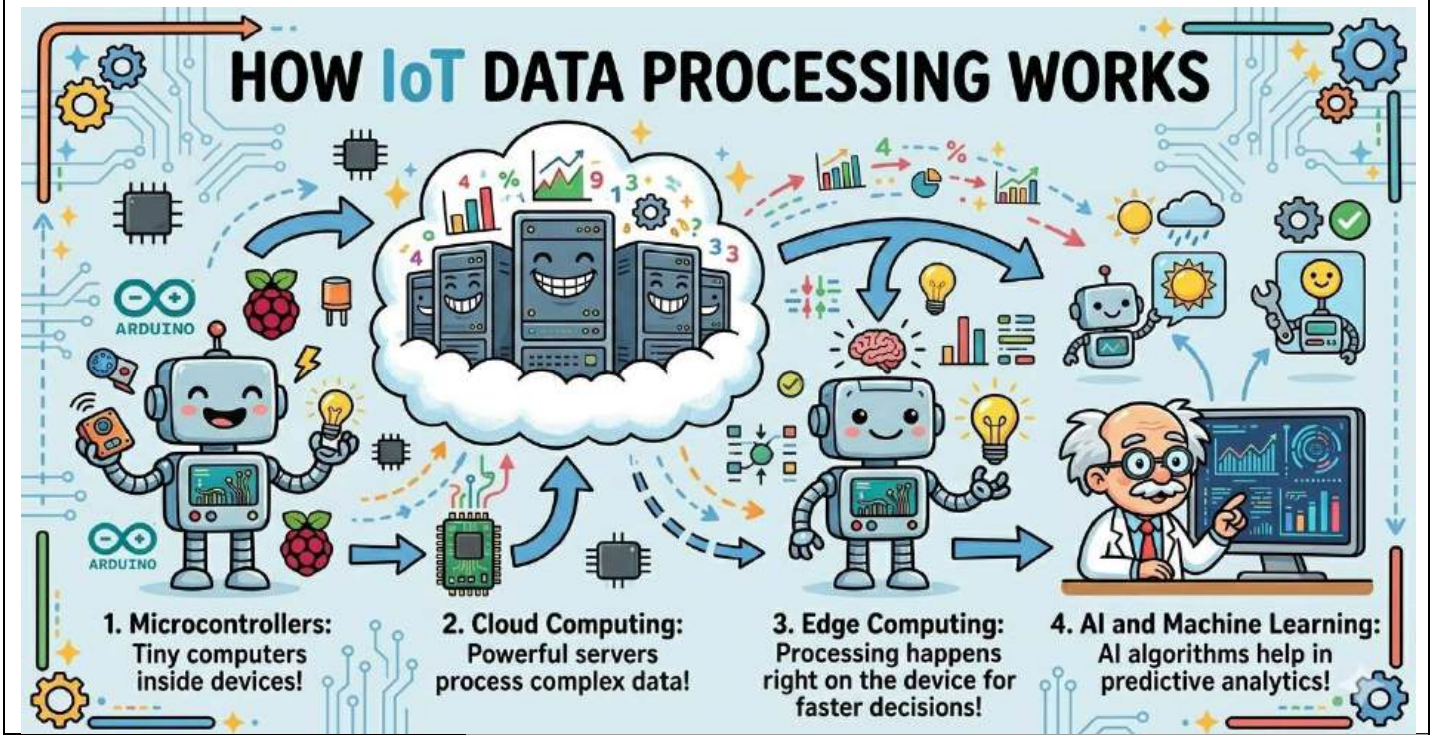
After data is transferred cloud or local server, the next step is data processing. This is where IoT transforms simple readings into meaningful insights.

What Happens During Data Processing?

- Data is analyzed in real time
- System identifies patterns and abnormalities
- Historical data is compared with current values
- Smart algorithms predict future trends
- Automatic triggers activate based on preset thresholds

How It Works:

- Microcontrollers, Tiny computers inside devices (like Arduino or Raspberry Pi)
- Cloud computing, Powerful servers process complex data
- Edge computing: Processing happens right on the device for faster decisions
- AI and Machine Learning: AI algorithms help in predictive analytics





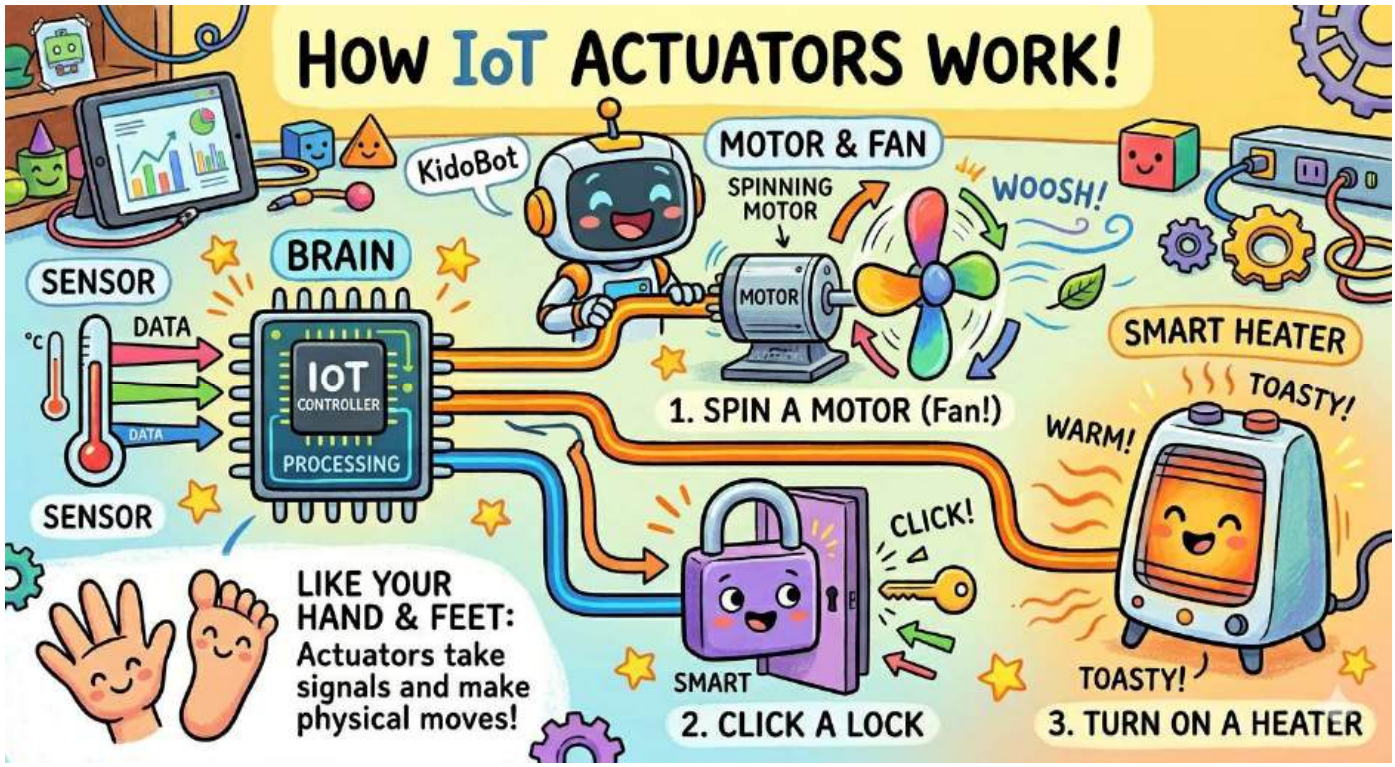
2.4 Actuators: Hands & Feet

Actuators are the components that perform physical actions, manipulate the environment, or create motion based on the data received from sensors and processed by a controller. They convert electrical signals into tangible, real-world actions

An IoT system can control one actuator or many at the same time. They're like your Hand and Feet - take the digital signals from the "brain" and change them back into physical movements, like spinning a motor, clicking a lock, or turning on a heater.

Key Actuators you Should Know:

- Motors: Servo motors or DC motors that create motion.
- Valves: Controlled flow of liquid or gas in agricultural or industrial systems.
- Relays: Switches that turn high-power devices (heaters, lights) on and off.
- Pneumatic/Hydraulic Cylinders: Used for pushing, pulling, or lifting, similar to limb movement.





2.5 User Interface: The Display

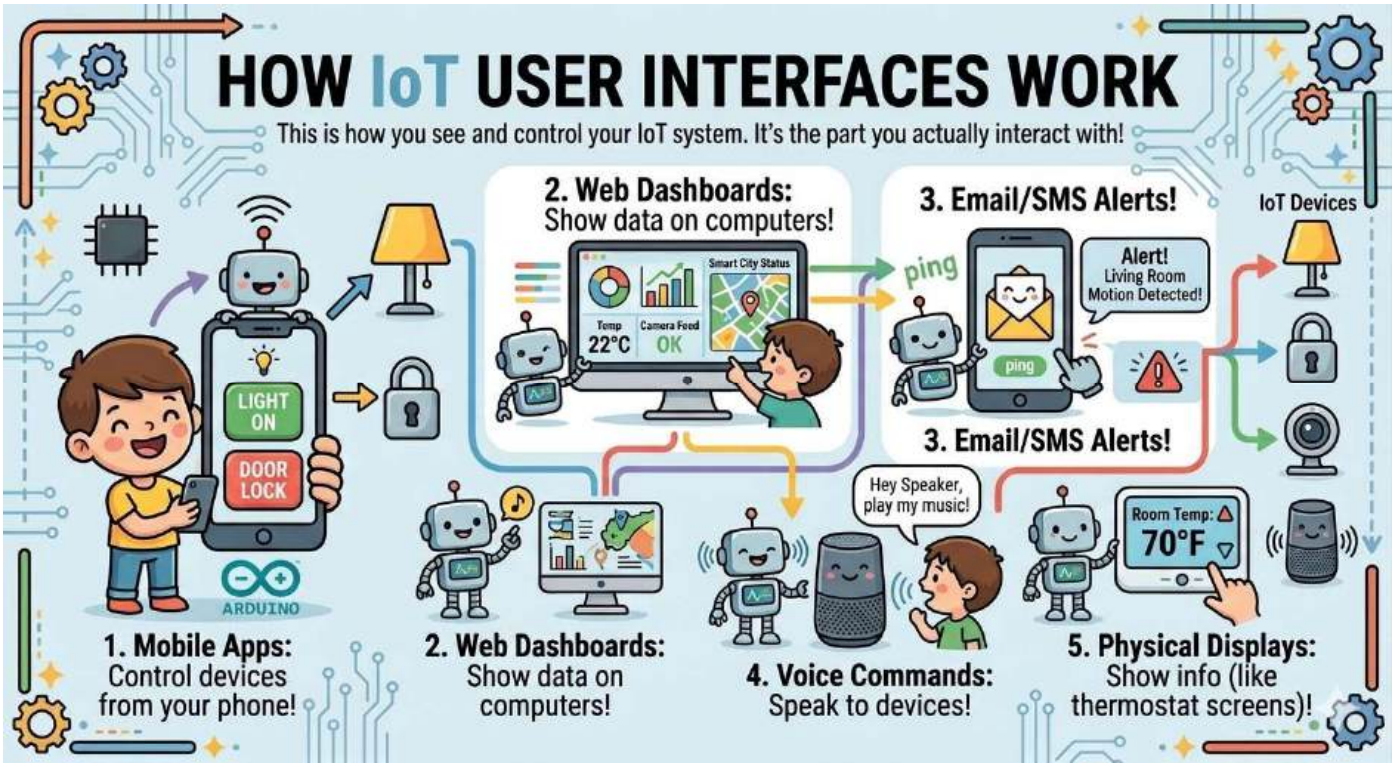
This is how you see and control your IoT system. It's the part you actually interact with—screens, apps, voice commands, or buttons.

Common Interfaces:

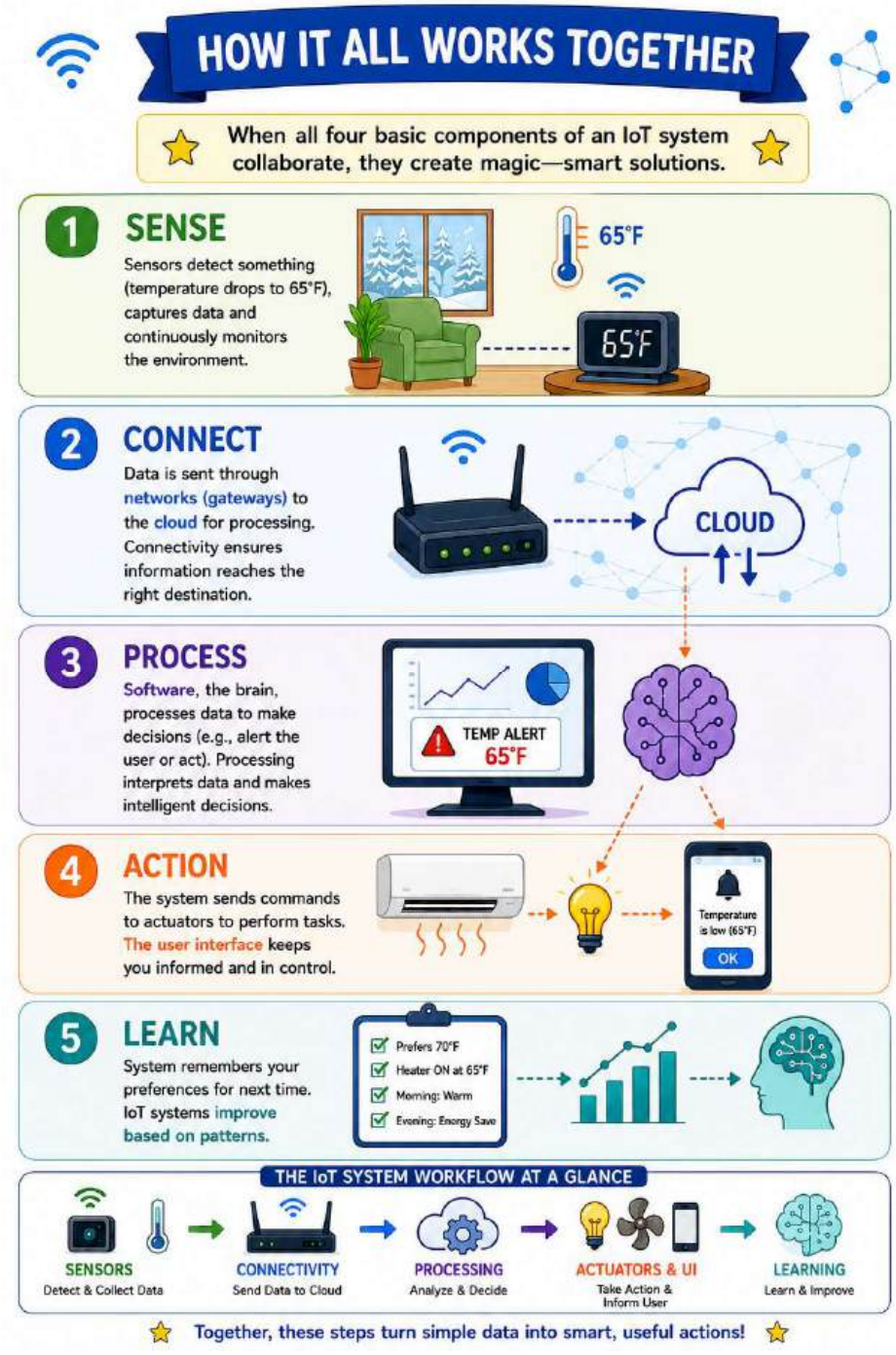
- Mobile Apps control devices from your phone
- Web Dashboards show data on computers
- Email/SMS Alerts
- Voice Commands let you speak to devices
- Physical Displays show information (like smart thermostat screens)

Why It's Important:

The user interface makes IoT accessible. Without it, you couldn't tell your system what to do or see what's happening.



2.6 How Do All IoT Components Work Together?





3. IoT with Arduino

3.1 What is Microcontroller?

A microcontroller is a tiny, super-smart computer chip inside everyday devices that tells them what to do. Think of it as a "brain on a chip," acting like a mini-computer that controls one specific job, such as making a toy robot move, turning on a toaster, or blinking the lights on a traffic signal.

What is the difference between Microcontroller and Computer?

- A microcontroller is a tiny, specialized computer on a single chip designed to control one specific task. They are used in embedded systems like in a microwave or washing machine.
- A computer (like a laptop) is a powerful, general-purpose system designed to multitask, run operating systems, and handle complex software. They are used for general computing (web browsing, gaming, editing).

3.2 Microcontrollers in IoT

IoT means connecting everyday objects to the internet so they can send information or be controlled remotely. Microcontrollers are the brains that make this possible.

They make devices "smart" by following a Sense → Think → Act loop:

- Sense: The controller reads data from sensors (e.g., "Is the room dark?", "Is the plant thirsty?").
- Think: The controller runs a program to decide what to do (e.g., "If dark, turn on the light").
- Act: The controller sends a command to an actuator (e.g., flip the switch).

Examples of Microcontrollers used in IoT

Students can use these popular microcontrollers to build their own smart devices:

- **Arduino Uno/Nano:**
  Great for beginners to learn basic IoT and robotics.

- **ESP32/ESP8266:**
  Powerful chips with built-in Wi-Fi and Bluetooth, perfect for connecting projects to the internet.

- **Raspberry Pi Pico:**
  A small, affordable, and fast board used in modern IoT projects.

3.3 What is Arduino?

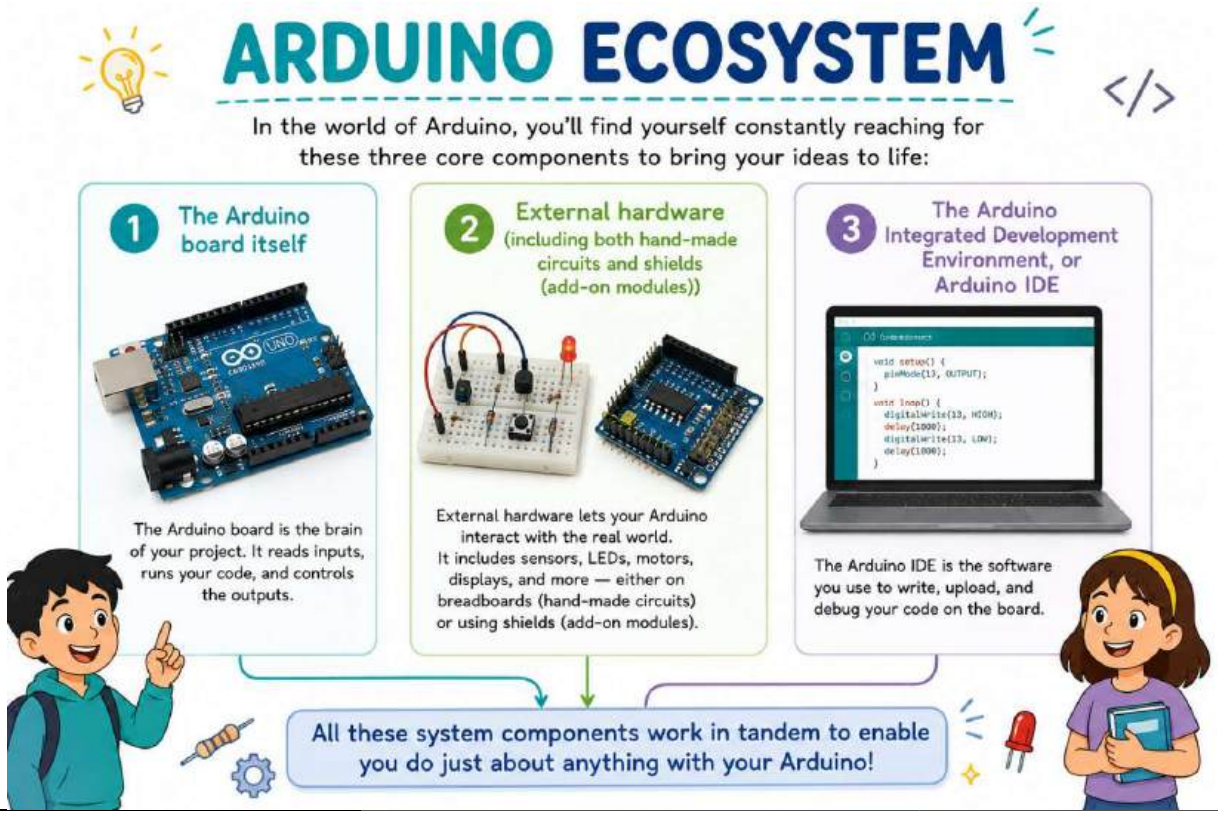
Arduino is an open-source electronics platform that combines simple hardware with easy-to-use software. At its core, an Arduino board is a small microcontroller that you can program using the Arduino integrated development environment (IDE).

By equipping the Arduino with sensors, actuators, lights, speakers, add-on modules (called shields), and other integrated circuits, you can turn the Arduino into a programmable "brain" for IoT systems or just about any control system.

Arduino Ecosystem

In the world of Arduino, you'll find yourself constantly reaching for these three core components to bring your ideas to life:

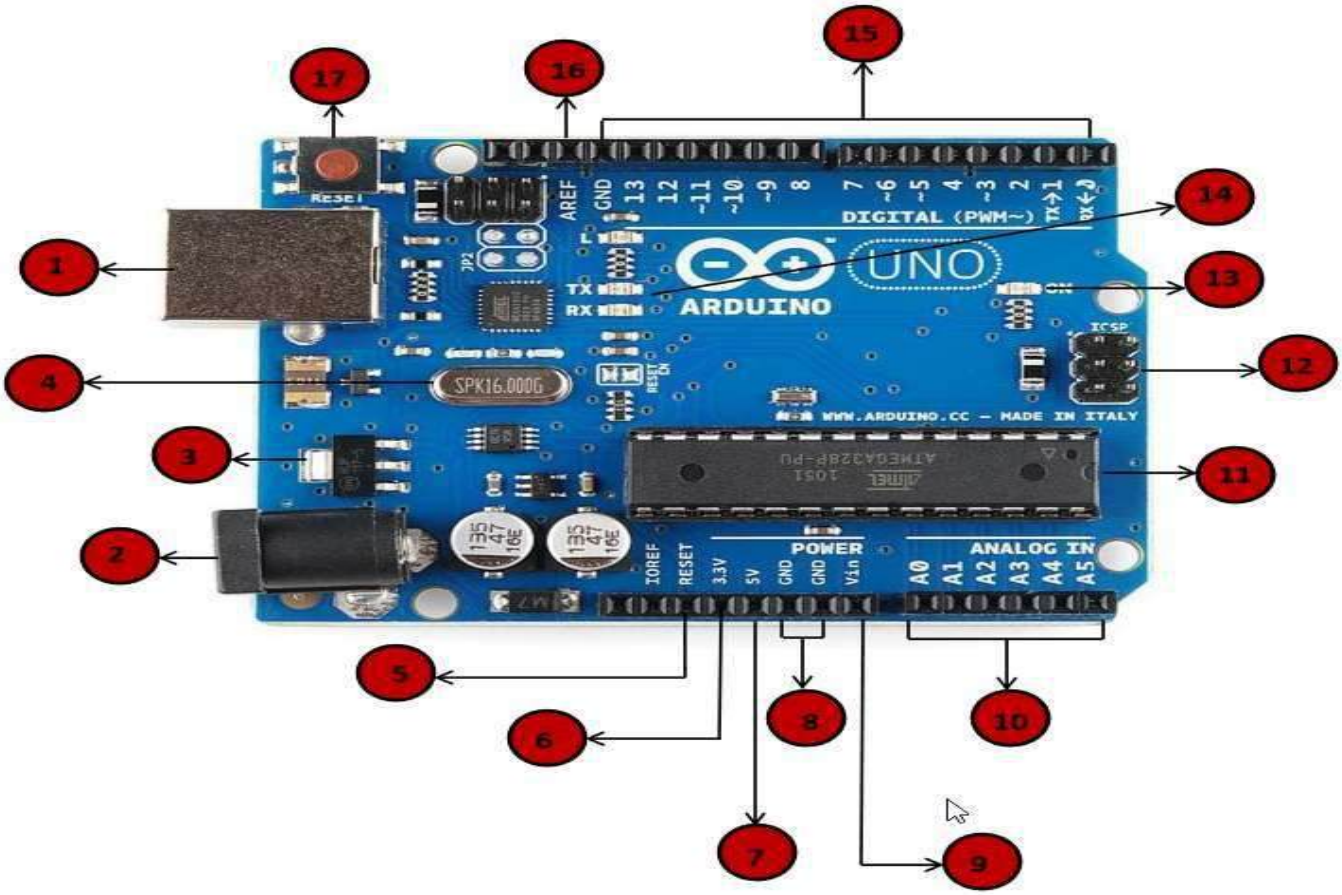
- The Arduino board itself
- External hardware (including both hand-made circuits and shields (add-on modules))
- The Arduino integrated development environment, or Arduino IDE



3.4 Arduino Uno Board

The Arduino Uno is one of the most popular and widely used Arduino boards. It is based on the ATmega328P microcontroller and provides an optimal balance of performance, features, and cost. This makes it suitable for a broad spectrum of applications, ranging from simple beginner projects to moderately complex systems.

Take a moment to examine the Arduino Uno Board:



1 → Power USB

Arduino board can be powered by using the USB cable from your computer. All you need to do is connect the USB cable to the USB connection (1).

- **2 → Power (Barrel Jack)**

  Arduino boards can be powered directly from the AC mains power supply by connecting it to the Barrel Jack (2).

- **3 → Voltage Regulator**
  The function of the voltage regulator is to control the voltage given to the Arduino board and stabilize the DC voltages used by the processor and other elements.

- **4 → Crystal Oscillator**
  The crystal oscillator helps Arduino in dealing with time issues. How does Arduino calculate time? The answer is, by using the crystal oscillator. The number printed on top of the Arduino crystal is 16.000H9H. It tells us that the frequency is 16,000,000 Hertz or 16 MHz.

- **5, 17 → Arduino Reset**
  You can reset your Arduino board, i.e., starts your program from the beginning. You can reset the UNO board in two ways. First, by using the reset button (17) on the board. Second, you can connect an external reset button to the Arduino pin labelled RESET (5).

- **6, 7, 8, 9 (3.3, 5, GND, Vin)**
  - 3.3V (6): Supply 3.3 output volt
  - 5V (7): Supply 5 output volt
  - GND (8): (Ground): There are several GND pins on the Arduino, any of which can be used to ground your circuit.
  - Vin (9): This pin also can be used to power the Arduino board from an external power source, like AC mains power supply.

    Most of the components used with Arduino board works fine with 3.3 volt and 5 volt.

- **10 → Analog pins**
  The Arduino UNO board has five analog input pins A0 through A5. These pins can read the signal from an analog sensor like the humidity sensor or temperature sensor and convert it into a digital value that can be read by the microprocessor.

- **11→ Main Microcontroller**
  - Each Arduino board has its own microcontroller (11).
  - You can assume it as the brain of your board.
  - The main IC (integrated circuit) on the Arduino is slightly different from board to board.

- **12 → ICSP pin**
  Mostly, ICSP (12) is an AVR, a tiny programming header for the Arduino consisting of MOSI, MISO, SCK, RESET, VCC, and GND. It is often referred to as an SPI (Serial Peripheral Interface), which could be considered as an "expansion" of the output. Actually, you are slaving the output device to the master of the SPI bus.

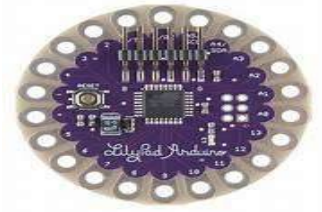
- **13 → Power LED indicator**
  This LED should light up when you plug your Arduino into a power source to indicate that your board is powered up correctly. If this light does not turn on, then there is something wrong with the connection.

- **14 TX and RX LEDs**
  On your board, you will find two labels: TX (Transmit) and RX (Receive). They appear in two places on the Arduino UNO board.
  - First, at the digital pins 0 and 1, to indicate the pins responsible for serial communication.
  - Second, the TX and RX led (13). The TX led flashes with different speed while sending the serial data. The speed of flashing depends on the baud rate used by the board. RX flashes during the receiving process.

- **15 → Digital I / O**
  - The Arduino UNO board has 14 digital I/O pins (15).
  - Out of which 6 provide PWM (Pulse Width Modulation) output.
  - These pins can be configured to work as input digital pins to read logic values (0 or 1) or as digital output pins to drive different modules like LEDs, Relays, etc. The pins labeled “~” can be used to generate PWM.

- **16 → AREF**
  AREF stands for Analog Reference. It is sometimes, used to set an external reference voltage (between 0 and 5 Volts) as the upper limit for the analog input pins.

3.5 Popular Arduino Boards

S.NO.	NAME OF BOARD	IDENTIFICATION	FUNCTION	APPLICATIONS
1	Arduino Nano		Used where small size and portability are needed	Wearables, mini robots, IoT devices, portable gadgets, DIY electronics
2	Arduino Mega 2560		Used in complex projects requiring many connections	3D printers, CNC machines, large robotics, data logging, industrial control
3	Arduino Lilypad		Used in smart clothing and textile projects	Smart clothing, fitness tracking, fashion tech, health monitoring
4	Arduino Leonardo		Used for automation and computer control tasks	Gaming controllers, HID devices, automation systems, custom keyboards

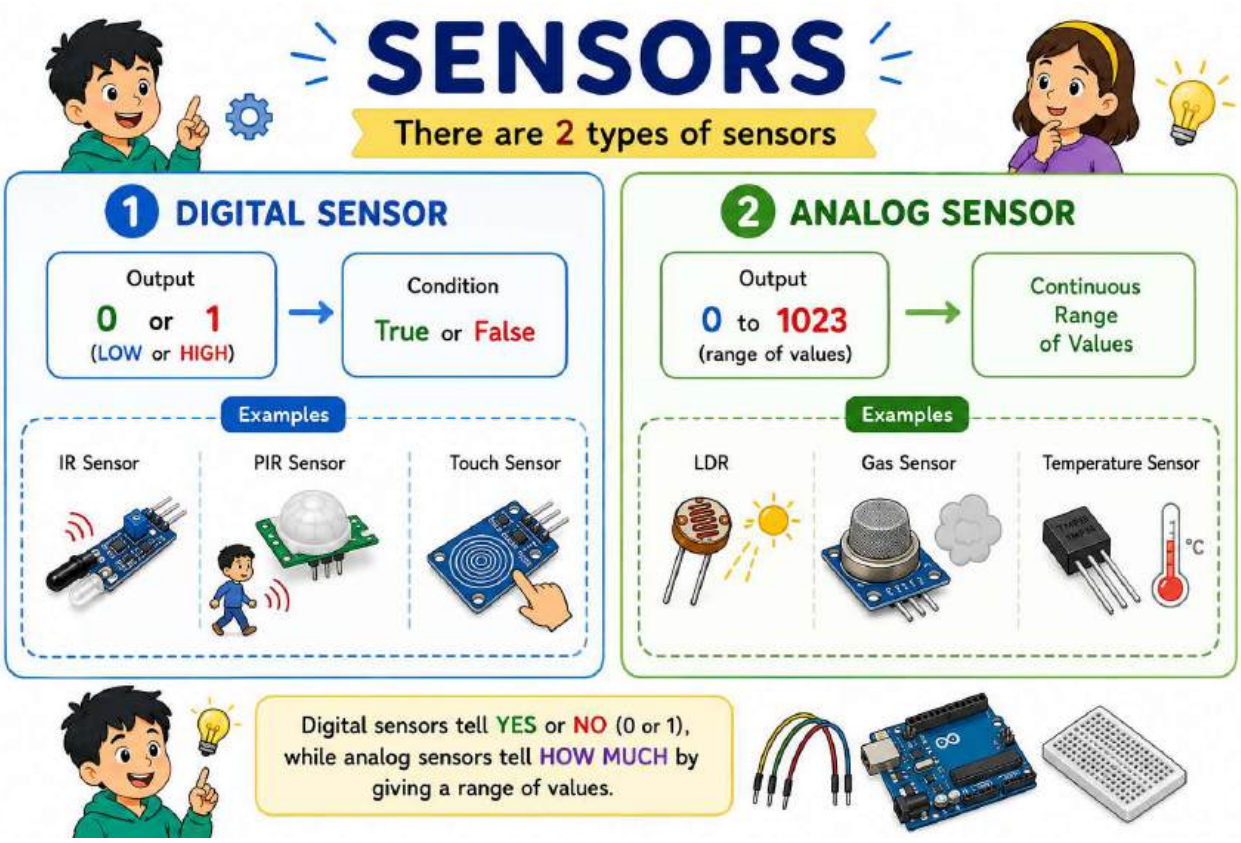
3.6 Electronic Components and Modules used with Arduino

3.6.1 SENSORS:

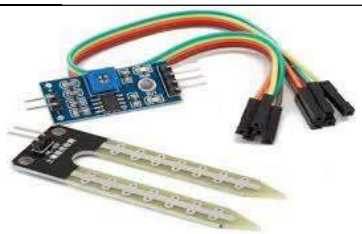
There are 2 types of sensors

- **Digital Sensor**
  - Digital sensors give discrete output, usually 0 or 1 (LOW or HIGH).
  - They indicate whether a condition is true or false.
    Examples: IR Sensor, PIR Sensor, Touch Sensor

- **Analog Sensor**
  - Analog sensors give a continuous range of values.
  - The output is not just 0 or 1, but can vary (e.g., 0 to 1023 in microcontrollers like Arduino).
  - Examples: Light Dependent Resistor (LDR), Gas Sensor, Temperature Sensor



S.NO.	NAME OF SENSOR	IDENTIFICATION	FEATURES	APPLICATIONS
1.	Capacitive Touch Sensor TTP223		Detects touch by change in capacitance	Touch switches, Doorbells,Mobile touch panels, Touch lamps, Elevator buttons
2.	Light Dependent Resistor (LDR)		Resistance varies with light intensity	Automatic street lights Light sensing ,Solar lights ,Camera control , Smart lighting
3.	IR Sensor		Uses infrared reflection to detect objects	Obstacle detection ,Line follower robots , Proximity sensing ,Automatic doors , Remote systems , Object counting
4.	PIR Sensor(Passive Infrared Radial Sensor) HC-SR501		Detects motion using infrared radiation	Motion detection , Security alarms,Auto lighting,Smart surveillance,Energy saving systems,Auto doors
5.	Ultrasonic Sensor (HC-SR04)		Uses sound waves to measure distance	Parking sensors ,Obstacle avoidance , Water level detection , Industrial automation

6.	Soil Moisture Sensor		Measures water content using conductivity	Smart irrigation, Auto watering,Agriculture monitoring ,Greenhouse control , Gardening
7.	Gas Sensor (MQ-2,MQ-4,MQ-135)		Detects gases by resistance change	Gas leakage detection, Air quality monitoring , Smoke detection , Pollution control
8.	Humidity andTemperature Sensor (DHT11)		Measures temperature & humidity digitally	weather monitoring, home automation, agriculture, industrial monitoring, Healthcare
9.	Rain Water Sensor		Detects presence of rainwater	Rain detection ,Auto wipers ,Irrigation control , Weather systems ,Flood alert
10.	Microphone sensor		Converts sound waves into electrical signals	Sound detection ,Voice recognition,Noise monitoring,Security systems,Voice control, Recording systems

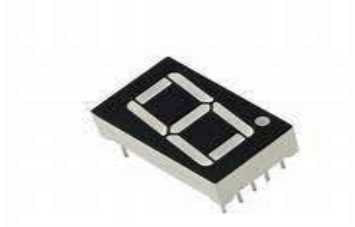
3.6.2 Actuators(Motors)

Actuators are components that convert electrical signals into mechanical output. They can be compared to human muscles, as muscles transform signals from the brain into physical movement, while actuators convert electrical signals into mechanical motion. Actuators are commonly implemented in the form of **motors**, which are widely used to produce movement in machines and electronic systems.

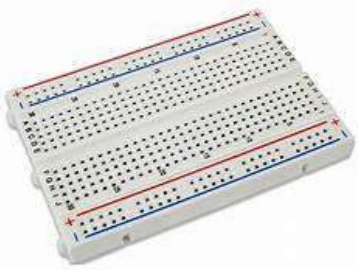
S.NO.	NAME OF MOTOR	IDENTIFICATION	KEY FEATURES	APPLICATIONS
1	BO Motor (Battery Operated Motor)		Low cost, lightweight, easy to use, battery operated	Toy cars , Line follower robots , DIY robotics projects , Small moving prototypes
2	DC Geared Motor		High torque, low speed, durable, suitable for heavy loads	Robotic vehicles , Conveyor belts , Automatic doors , Industrial machines,
3	Stepper Motor		Precise control, step-by-step rotation, good positioning	3D printers, CNC machines ,Camera positioning systems ,Robotics requiring precise control
4	Servo Motor		Controlled rotation with fixed angle (0°–180°or 0-360°)	Robotic arms , RC cars and airplanes ,Drone control systems, Automation systems
5	Vibration Motor		Compact size, produces vibration, low power	Mobile phone vibration, Game controllers, Alert systems , Wearable devices
6	Water pump module		Motor connected to a pump	Irrigation systems, Aquarium water circulation, Water fountains, Cooling systems

3.6.3 Output Devices

Output devices in Microcontroller are components that receive signals from the controller and convert them Display,sound, or movement.

S.NO.	NAME OF OUTPUT DEVICE	IDENTIFICATION	KEY FEATURES	APPLICATIONS
1	LED (Light Emitting Diode)		Used as status indicator (ON/OFF)	Indicator lights, traffic signals, decorative lighting, emergency signals,
2	LCD Display (16x2)		Used to show messages, sensor values, system data	Digital clocks, calculators, Arduino projects display
3	Buzzer		Used for alerts, warnings, notifications	Alarms, doorbells, timers, warning system security alerts
4	Relay Module		Used to turn ON/OFF appliances safely	Home automation, industrial control, smart switches, lighting systems, motor control
5.	7-Segment Display		Used to display numbers (0–9)	Counters, digital clocks, scoreboards, elevators, measuring instruments

3.6.4 OTHER IMPORTANT COMPONENTS

S.NO.	NAME OF COMPONENT	IDENTIFICATION	FUNCTION	APPLICATIONS
1	Breadboard		Used to build and test circuits without soldering	Circuit prototyping, testing
2	Resistor		Used to control current flow in the circuit	LED circuits, voltage control
3	Potentiometer		Used to vary input values manually	Volume control, brightness control
4	Power Supply (Battery/Adapter) 12volt		Used to power the system	All type of projects
5	Push Button		Used to give manual input	Remote,microwave panel switch

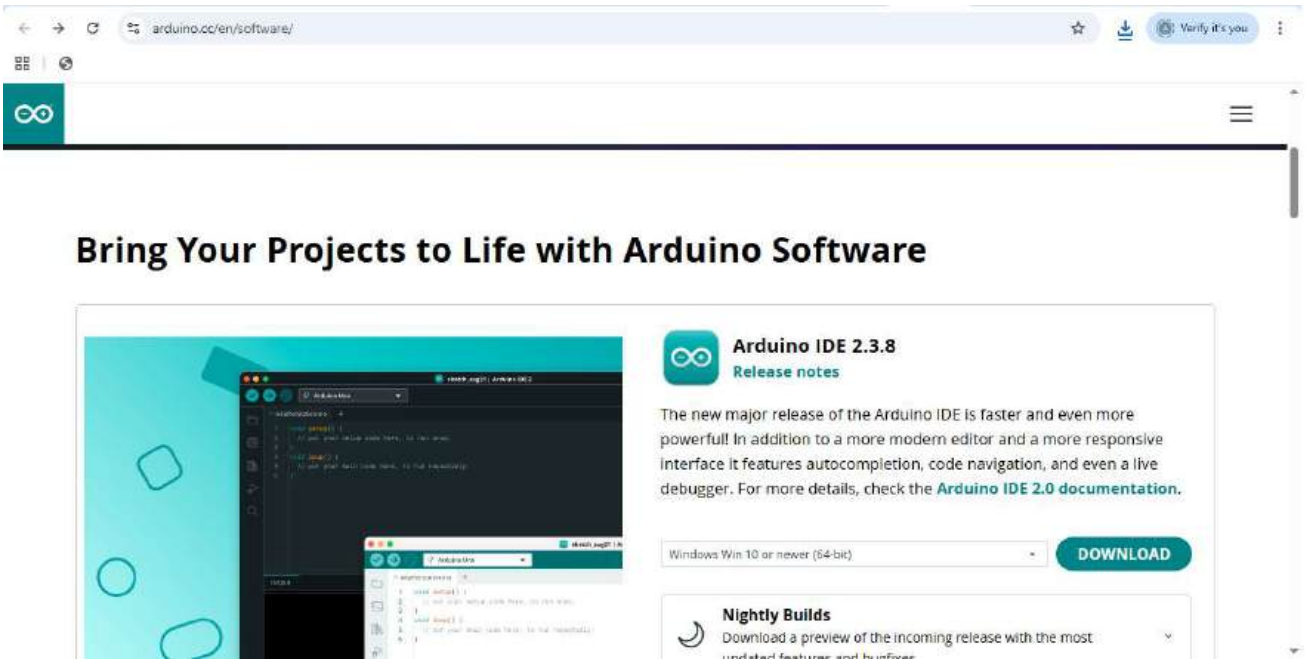
3.7 Getting Ready for Arduino Programming

- **What is Arduino IDE ?**

  The Arduino IDE (Integrated Development Environment) is a program you install on your computer. Think of it as a digital notebook where you write your instructions (code) for the robot.
  - **What it does:** It lets you write, check, and send your code to the Arduino board.
  - **How it works:** You connect your Arduino board to your computer with a USB cable, and the IDE sends the instructions through that cable.

- **Downloading and Installing the Arduino IDE**

  Access the Arduino website at www.arduino.cc and download the newest version of the IDE from the Download page.



After completing the download, unzip it. Inside, you'll find the Arduino IDE. New versions of the Windows IDE are available as an installer that you can download and run, instead of downloading a ZIP file.

➢ **Running the IDE and Connecting to the Arduino**

Now that you have the IDE downloaded and ready to run, you can connect the Arduino to your computer via USB, as shown



Figure : Arduino Uno connected to a computer via USB

Now, launch the Arduino IDE. You're ready to load your first program onto your Arduino.

Before you load the program, you need to tell the IDE what kind of Arduino you have connected and what port it is connected to.

- Go to Tools → Board and ensure that the right board is selected.
- Go to Tools → Serial Port and select the appropriate port.



3.8 Basics of Arduino Programming

Arduino programming is a fun way to bring electronics to life by telling a small computer (the board) what to do. The language is based on C++, which is a widely used, powerful coding language, but Arduino makes it simpler for beginners by using specific commands

➢ **What is an Arduino "Sketch"?**

- In Arduino, a program is called a Sketch. It's a set of instructions written in the Arduino IDE (Integrated Development Environment) on a computer.
- You write the sketch on your computer and upload it via USB to the Arduino board, which then remembers it and runs it
- The file extension is **.ino**

➢ **Two Main Functions**

Every Sketch must have two main parts:

- **setup() : Runs once when Arduino starts**

  This runs only once when the Arduino turns on. You use this to set up things, like telling the Arduino if a pin is an input (sensor) or output (light/motor).

```
void setup() {
  pinMode(13, OUTPUT);   // Tell pin 13 to be an output
  Serial.begin(9600);    // Start talking to the computer
}
```

- **loop(): Runs forever on repeat**

  This runs over and over again forever, until you turn the power off. This is where your main logic goes (e.g., blink light, read sensor)

```
void loop() {
  digitalWrite(13, HIGH); // Turn LED on
  delay(1000);            // Wait 1 second
  digitalWrite(13, LOW);  // Turn LED off
  delay(1000);            // Wait 1 second
}
```

The code above blinks an LED forever.



- **Key Functions You Must Know**

Command	What it does	Example
`pinMode(pin, MODE)`	Sets a pin as INPUT or OUTPUT	`pinMode(3, OUTPUT)`
`digitalWrite(pin, value)`	Turns a pin ON HIGH or OFF LOW	`digitalWrite(3, HIGH)`
`digitalRead(pin)`	Checks if a pin is ON or OFF	`if(digitalRead(2) == HIGH)`
`analogRead(pin)`	Reads a sensor value from 0 to	`1023int light = analogRead(A0)`
`analogWrite(pin, value)`	Sends power from 0 to 255 for dimming/fade	`analogWrite(9, 128)`
`delay(time)`	Pauses the program in milliseconds	`delay(1000) = 1 sec pause`
`serial.println(data)`	Sends data to your computer screen	`Serial.println("Hello")`

- **The Core Coding Basics**
  - **Comments // or /* */:**
    Text that the computer ignores. Use this to take notes.
  - **Semicolon (;) :**
    Every command must end with a semicolon, like a period at the end of a sentence. Arduino gets confused without it
  - **Braces { }:**
    Used to group code together (e.g., inside setup or loop)
  - **Case matters:**
    pinMode works, but PinMode will give error.

- **Variables**

Variables store info Arduino needs to remember.

```
int buttonState = 0;      // Whole numbers: -32768 to 32767
float temperature = 25.5; // Numbers with decimals
bool isOn = true;         // Only true or false
```

Put them before setup() if you want to use them everywhere.



- **If Statements: Making Decisions**

Arduino can choose what to do based on conditions:

```
int sensorValue = analogRead(A0);

if(sensorValue > 500) {
  digitalWrite(13, HIGH);  // If bright, turn LED on
} else {
  digitalWrite(13, LOW);   // Otherwise, turn LED off
}
```

- **Loops: Repeating Stuff**

Besides the main loop(), you can use for loops to repeat actions:

```
for(int i = 0; i < 5; i++) {
  digitalWrite(13, HIGH);
  delay(200);
  digitalWrite(13, LOW);
  delay(200);
}  // This blinks LED 5 times fast
```

- **Libraries**

Sometimes you want to do something complicated, like control a tiny display screen. Instead of writing 100 lines of code, you can use a Library.

- **What it is:** A pre-written file of code created by someone else that makes hard tasks easy.
- **Analogy:** It's like using a LEGO instruction booklet to build a car, instead of figuring out how to design the car yourself!



3.9 Arduino Example Program: Blink

The Blink program is usually the first thing students try. It makes an LED turn on and off over and over. Simple, but it teaches how Arduino works.

- **What You Need?**
  - Arduino Uno Board
  - USB Cable (to connect Arduino to your computer)
  - Arduino IDE Software (installed on your computer)

- **The Circuit (No Breadboard Needed!)**
  Most Arduino Uno boards have a tiny LED labeled 'L' connected directly to Pin 13. So, for the basic Blink, you don't need to plug in any external wires or LEDs!

- **The Full Code**

```
// This program blinks the built-in LED

void setup() {
  pinMode(13, OUTPUT);    // Set pin 13 to control things
}

void loop() {
  digitalWrite(13, HIGH); // Turn the LED ON
  delay(1000);            // Wait 1 second
  digitalWrite(13, LOW);  // Turn the LED OFF
  delay(1000);            // Wait 1 second
}
```

- **The Code Explained**
  - **void setup() { ... } Runs once when Arduino starts**
    - **`pinMode(13, OUTPUT);`**
      - Pin 13 has a tiny LED already on the Arduino board.
      - This line says "Hey pin 13, your job is to send power out, not read it in".
      - OUTPUT means Arduino controls the pin. INPUT would mean it listens



- **void loop() { ... } Runs forever on repeat**
  - **`digitalWrite(13, HIGH);`**
    - HIGH means "send 5V". This turns the LED on.
    - digitalWrite only does ON HIGH or OFF LOW. Nothing in between.
  - **`delay(1000);`**
    - Pause for 1000 milliseconds. That equals 1 second.
    - Without delays, the LED would blink so fast your eyes couldn't see it.
  - **`digitalWrite(13, LOW);`**
    - LOW means "send 0V". This turns the LED off.
  - **`delay(1000); // again`**
    - Wait another second with the LED off, then loop() repeats.

    .

➢ **How to Run It?**

- Connect your Arduino to your computer using the USB cable.
- In the Arduino IDE, go to Tools > Board and select "Arduino Uno".
- Go to Tools > Port and select the port your board is connected to.
- Click the Upload button (the arrow icon in the top left).
- Watch the LED on your board blink!

➢ **Fun Experiments to Try**

- **Change the speed:**
  Make both delays delay(100). Now it blinks super fast.
- **Make it uneven:**
  Use delay(2000) for ON and delay(200) for OFF. Now it's a long blink, short pause.
- **Use a variable:**
  Add int waitTime = 500; at the top, then use delay(waitTime). Now you can change speed by editing one number.



3.10 Arduino Project: Motion Sensor with LED Light

- **Introduction to Tinkercad Simulation Software**

  Arduino Tinkercad is a free online simulation and 3D design software developed by Autodesk. It is widely used by students, teachers, beginners, and makers for learning electronics, coding, circuit design, and 3D modeling in an easy and interactive way. Tinkercad works directly in a web browser, so there is no need to install any special software on the computer.

One of the most useful features of Tinkercad is its **Circuits Simulation** environment. It allows users to create virtual electronic circuits using components such as LEDs, resistors, sensors, buzzers, motors, switches, and displays. Users can connect these components with virtual wires and test the circuit safely before making it in real life. This saves time, reduces cost, and avoids damage to electronic components.

Tinkercad also supports programming of microcontrollers like the Arduino Uno. Students can write programs using both **Block Coding** and **Text Coding** in the Arduino language. The simulation runs instantly, helping learners understand how the hardware and software work together. It is very useful for learning concepts related to robotics, automation, Internet of Things (IoT), and embedded systems.

Another important feature is the **3D Design Tool**, which helps users create 3D models and objects. These models can be used for educational projects and can also be exported for 3D printing. The simple drag-and-drop interface makes designing easy even for beginners.



➢ **Motion Sensor with LED Light using Tinkercad**

We will create an automatic lighting system using a PIR motion sensor and LED in Tinkercad. The LED turns ON when motion is detected and turns OFF when no motion is present.

- **Components Required**
  - Arduino Uno R3
  - PIR Motion Sensor
  - LED
  - 220 Ohm Resistor
  - Breadboard
  - Jumper wires.

- **Working Principle**

  A PIR (Passive Infrared) Motion Sensor detects infrared radiation from humans or moving objects. When motion is detected, the sensor sends a HIGH signal to the Arduino. The Arduino then turns ON the LED automatically.

  This project is commonly used in:
  - Automatic room lights
  - Security systems
  - Smart home automation
  - Energy-saving systems

- **Circuit Connections**

Component	Arduino Pin
PIR VCC	5V
PIR GND	GND
PIR OUT	Digital Pin 2
LED Positive (+)	Digital Pin 13
LED Negative (-)	GND through 220Ω resistor



Step-by-Step Procedure:

- **Step 1: Open Tinkercad**
  1. Go to Tinkercad

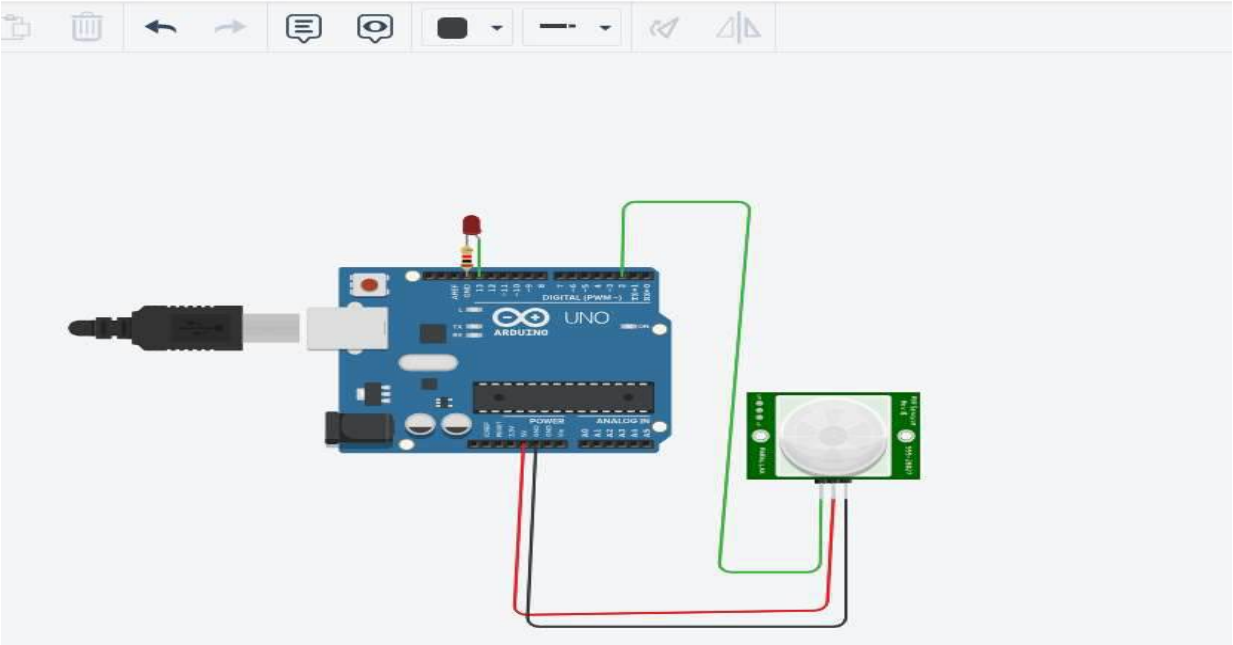
  2. Click on Login and join with personal account

3. Add email id information: gmail account or apple account

4. Click on create and select Circuits

- **Step 2: Add Components**
  Drag and drop following components into the workspace:
  - Arduino Uno
  - Breadboard
  - PIR Motion Sensor
  - LED
  - Resistor
  - Jumper wires

- **Step 3: Connect PIR Sensor**
  - Connect VCC pin to 5V of Arduino.
  - Connect GND pin to GND of Arduino.
  - Connect OUT pin to Digital Pin 2.



- **Step 4: Connect LED**
  - Connect LED positive leg (long leg) to Pin 13.
  - Connect LED negative leg to 220Ω resistor.
  - Connect resistor to GND.

- **Step 5: Select Code and Text**

- **Step 6: Write Arduino Code**

```
      int pirPin = 2;
      int ledPin = 13;
      int motion;

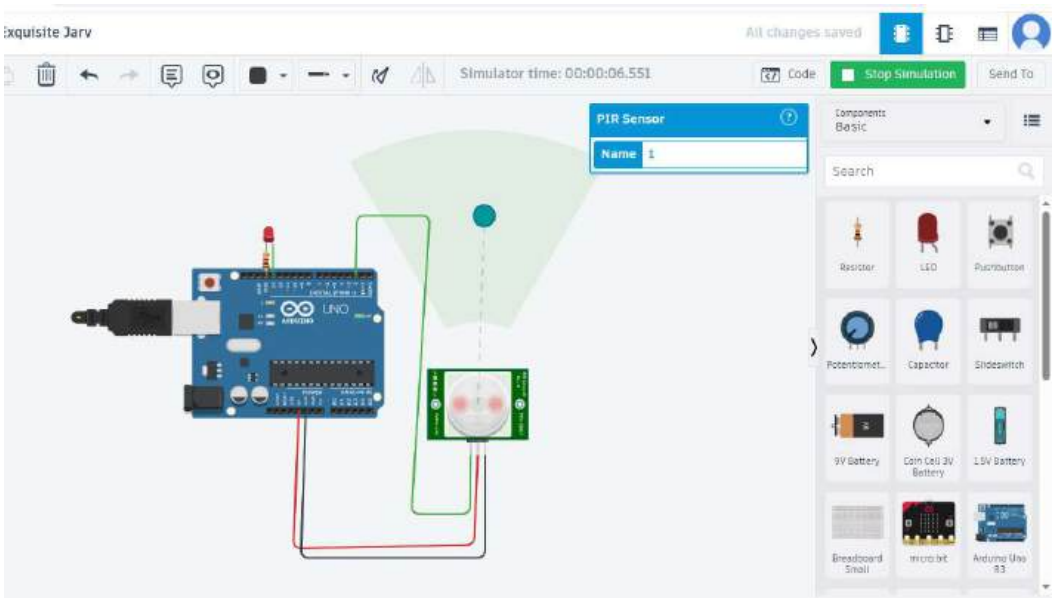
      void setup()
      {
        pinMode(pirPin, INPUT);
        pinMode(ledPin, OUTPUT);

        Serial.begin(9600);
      }

      void loop()
      {
        motion = digitalRead(pirPin);

        if (motion == HIGH)
        {
          digitalWrite(ledPin, HIGH);
          Serial.println("Motion Detected!");
        }
        else
        {
          digitalWrite(ledPin, LOW);
          Serial.println("No Motion");
        }
}
```

- **Step 7: Start Simulation**
  - Click Start Simulation.
  - Change the PIR sensor state to simulate motion.
  - Observe the LED:
    - Motion detected → LED ON
    - No motion → LED OFF



- **Output**
  - LED glows automatically when motion is detected.
  - LED turns OFF when there is no movement.



4. Practice Questions

4.1 Practice Questions: IoT Basics

1. IoT stands for what? a) Internet of Tech b) Intelligent of Things c) Internet of Things d) Incorporation of Technology	2. IoT is based on _____ technology. a) Hardware b) Software c) None d) Both of these
3. What is an IoT Network? a) A collection of networked devices. b) A collection of Interconnected devices. c) A collection of signalled devices. d) None of the above	4. Which application area uses IoT to improve energy efficiency? a) Smart Home b) Smart Farming c) Smart Healthcare d) All of the above
5. When was the term “Internet of Things” coined? a) 1998 b) 1999 c) 2000 d) 2002	6. IIoT stands for what? a) Industrial Internet of Things b) Indepth Internet of Things c) Innovative Internet of Things d) Information Internet of Things
7. Which problem can IoT solve best? a) Forgetting Things b) Monitoring Health c) Saving Energy d) All of the above	8. IoT sensors detect unusual vibrations in equipment and alert technicians before failure. What is this application? a) Reactive Maintenance b) Predictive Maintenance c) Manual Repair d) Scheduled Maintenance
9. An IoT system not only records attendance but predicts student performance trends. This combines IoT with: a) Basic Networking b) Data Analytics / AI c) Manual Systems d) Hardware Repair	10. If School Bus GPS fails but the system continues tracking using alternative sensors, what is demonstrated? a) Scalability b) Fault Tolerance c) Virtualization d) Compression



11. The storage is _____ in IoT. a) Limited b) Unlimited c) Not available d) All of these	12. In IoT, what does the term "gateway" refer to?. a) A device that connects IoT devices to the internet b) A physical entrance to a building c) An IoT sensor d) A data center
13. Which IoT connectivity technology is known for its low power consumption and is commonly used in wearable devices? a) Wi-Fi b) Zigbee c) Bluetooth d) NFC	14. The ..... is a way to connecting electronic components to each other without having solder them together. a) Servo Motor b) Arduino c) GPU d) Breadboard
15. What is the primary benefit of integrating cloud computing with IoT systems? a) On-demand scalability and storage b) Increased power consumption c) Reduced sensor accuracy d) Improved physical security	16. What is the first step in an IoT system? a) Processing b) Action c) Sensing d) Display
17. What can IoT systems do automatically? a) Make decisions b) Take actions based on data c) Control devices d) All of the above	18. What is meant by “real-time data” in IoT? a) Old stored data b) Data received instantly or quickly c) Data not used d) Data deleted
19. What is the purpose of an actuator? a) Collect data b) Store data c) Perform action (like turning ON motor) d) Display data	20. What will happen if sensors give incorrect data? a) System works perfectly b) Wrong decisions may be taken c) No effect d) Faster processing



21. What is latency in IoT? a) Device size b) Delay in data transmission c) Type of sensor d) Storage capacity	22. What is the purpose of an actuator? a) Collect data b) Store data c) Perform action (like turning ON motor) d) Display data
23. What is the final step in a typical IoT system? a) Data sensing b) Data transmission c) Action or decision d) Data collection	24. In IoT, what does “edge computing” primarily reduce? a) Device cost b) Latency by processing data near the source c) Number of sensors needed d) WiFi password requirements
25. What security risk is unique to IoT compared to regular IT systems? a) Phishing emails b) Physical tampering of distributed devices c) Slow internet d) Software updates	26. Which wireless technology is best for a smart home bulb 10 meters from the router? a) LoRaWAN b) Satellite c) WiFi or Bluetooth d) Ethernet cable
27. Why would an IoT farm sensor in a remote field use LoRa instead of WiFi? a) LoRa is faster than WiFi b) LoRa works for many kilometers with very little power c) LoRa can stream YouTube d) WiFi does not work with plants	28. What is the primary reason for adopting "low-power" designs in IoT devices? a) To increase data throughput b) To enable battery operation for extended periods c) To reduce the cost of the microcontroller d) To eliminate the need for sensors
29. Which state consumes the highest power in a typical IoT sensor node? a) Sleep state b) Data processing (computing) c) Active transmitting/radio communication d) Data sampling	30. In environmental monitoring, IoT sensors are used to: a) Play music b) Measure pollution levels c) Charge devices d) Display images



4.2 Practice Questions: Arduino

31. The Arduino board is also called a _____? a) Microprocessor b) Timer c) Oscillator d) Microcontroller	32. Full form of IDE is_____. a) Integrated Develop Expansion b) Integrated Development Environment c) Interleaving Deficit Expander d) Interfaced Developed Environment
33. _____ is the language used by the Arduino IDE to code a) C b) C++ c) Java d) Python	34. Full form of LED is_____ a) Laser Emit Diode b) Light Emitting Diode c) Light Emitter Diode d) None of the above
35. To ground the circuit _____ pin is used in Arduino a) vin b) b) Reset c) c) Gnd d) d) Tx and Rx	36. How many Analog pins exist in Arduino? a) 5 b) 6 c) 9 d) 12
37. The Arduino Programming structure consist of? a) setup() and loop() b) loop() and start() c) function() and loop() d) setup() and fuction()	38. In the setup() function code is executed a) two time b) one time c) continuously d) none of the above
39. Serial.begin() function is used for_____. a) Serial Communication b) LOOP condition c) Communication d) Monitoring	40. Variables in Programming are used to _____. a) store data b) show data c) collection d) clear data



41. Reset button in Arduino Board used to_____. a) erase the code b) Restart the program c) clear all variables d) stop execution	42. delay(10000) function is used in Arduino code for giving_____. a) 10second delay b) 100 second delay c) 1 second delay d) 0.5 second delay

43. Which of the following is syntax for setting the mode of digital pin as input in Arduino sketch?
a) pinMode(pin,LOW);
b) pinMode(pin,HIGH);
c) pinMode(pin,INPUT);
d) pinMode(pin,OUTPUT);
	44. What is the maximum voltage Arduino UNO Board can handle?
	a) 5V
	b) 9V
	c) 12V
	d) 20V

45. What is the range of values returned by analogRead() in Arduino? a) 0 – 255 b) 0 - 1023 c) 0 - 511 d) 1 - 1024	46. Which sensor uses sound waves to measure the distance of an object? a) IR Sensor b) PIR Sensor c) Ultrasonic Sensor (HC-SR04) d) LDR
47. What are the two signal pins on an ultrasonic sensor called? a) VCC and GND b) Trigger and Echo c) Signal and Input d) Start and Stop	48. Which sensor is best used for a smart irrigation system to detect the moisture in soil? a) DHT11 b) MQ-2 c) Soil Moisture Sensor d) PIR Sensor
49. What is the full form of LDR, which is used to measure light intensity? a) Light Digital Resistor b) Low Density Resistor c) Light Dependent Resistor d) Linear Data Receiver	50. Which sensor is used to detect motion, commonly used in security systems? a) MQ-2 b) PIR Sensor c) LDR d) Ultrasonic Sensor



51. What type of sensor is the MQ-series (e.g., MQ-2, MQ-7)? a) Temperature sensor b) Gas/Smoke sensor c) Sound sensor d) Motion sensor	52. If an analog sensor is producing a voltage, and you want to detect when it crosses a certain threshold, what function is most useful? a) digitalWrite() b) analogRead() c) delay() d) analogWrite()
53. What is the output of a digital sensor? a) 0 to 1023 b) b) Continuous values c) c) Only 0 or 1 d) d) Voltage range	54. Which motor is commonly used for precise angle control? a) DC motor b) Servo motor c) Stepper motor d) Induction motor
55. Which motor requires a motor driver (like L298N) with Arduino? a) Servo motor b) DC motor c) Both A and B d) None	56. Stepper motor moves in: a) Continuous rotation b) Random motion c) Fixed steps d) Only clockwise direction
57. Which sensor detects human presence or motion? a) PIR sensor b) LDR c) Gas sensor d) Ultrasonic sensor	58. Why is an external power supply recommended for motors instead of Arduino 5V pin? a) Arduino cannot generate voltage b) Motors require high current which Arduino cannot supply c) Motors work only on AC d) Arduino pins give analog output only
59. What will happen if a DC motor is directly connected to Arduino pin? a) Motor will run normally b) Arduino pin may get damaged c) Motor will run faster d) Nothing will happen	60. In an ultrasonic sensor, distance is calculated using: a) Light intensity b) Sound wave reflection time c) Temperature change d) Resistance



61. A student designs a street light system using an LDR and Arduino. The light should turn ON only at night. What should be the correct logic?
    a) Light ON when LDR value is high
    b) Light ON when LDR value is low
    c) Light always ON
    d) Light ON only in daytime

62. A sensor gives fluctuating readings in a water tank project. What is the best approach?
    a) Add delay and averaging in code
    b) Ignore readings
    c) Remove sensor
    d) Increase voltage

63. A soil moisture sensor is used, but the motor (pump) runs continuously. What is the likely mistake?
    a) Wrong condition in code
    b) Sensor measures light
    c) Motor is broken
    d) Arduino has no power

64. Which Arduino board is used for clothing and textile projects?
    a) Arduino uno
    b) Arduino Mega
    c) Arduino Lilypad
    d) Arduino nano

65. Identify component.



    a) buzzer
    b) LED
    c) Vibrating Motor
    d) Water pump

66. For making traffic light project which type of display can be used?
    a) Seven segment Display
    b) LCD display
    c) both a and b
    d) None of the above

67. _____is used for control current flow in circuit.
    a) Resistor
    b) Capacitor
    c) LED
    d) all of above

68. Which type of switch is used for microwave panels?
    a) push button
    b) on/off switch
    c) both a and b
    d) tactile switch

69. Every command in program must end with ____
    a) :
    b) ;
    c) //
    d) .

70. LOW in programming indicates_____
    a) 0
    b) 1
    c) 10
    d) 11



71. Which component is commonly used to display text and numbers with Arduino? a) Motor b) LED c) LCD Display d) Sensor	72. What does LCD stand for? a) Light Control Device b) Liquid Crystal Display c) Light Crystal Display d) Liquid Control Device
73. Which Arduino library is commonly used for LCD displays? a) Servo.h b) Wire.h c) LiquidCrystal.h d) Display.h	74. What is the purpose of the potentiometer connected with an LCD? a) To increase speed b) To control brightness/contrast c) To store data d) To power the Arduino
75. Which communication protocol is commonly used with OLED displays and Arduino? a) HDMI b) USB c) I2C d) VGA	76. Which pin on the sensor is used for power supply? a) VCC b) GND c) RS d) EN
77. if statement is used for: a) Looping b) Decision making c) Delay d) Display	78. Which display is thinner and brighter compared to LCD? a) CRT b) OLED c) LED Bulb d) Speaker
79. Which type of display can show graphics and images easily? a) Seven Segment Display b) OLED Display c) Buzzer d) Relay	80. A seven-segment display is mainly used to display: a) Images b) Videos c) Numbers d) Sound



81. Which component helps Arduino send information to the display? a) Output pins b) Wheels c) Battery only d) Switch	82. What is the main purpose of connecting a display to Arduino? a) To increase voltage b) To show output data and messages c) To reduce memory d) To cool the circuit
83. Which display type uses less power? a) OLED b) CRT c) Projector d) TV Monitor	84. Which of the following is an example of an output device used with Arduino? a) Temperature Sensor b) Push Button c) LCD Display d) LDR Sensor
85. Identify this component  a) Motion Sensor b) Ultrasonic Sensor c) Infrared Sensor d) Soil moisture Sensor	86. Which function executes first when a sketch is built? a) set up() b) loop() c) void() d) loop()

87. Which code is correct for making pin 13 OUTPUT?
a) pinMode(13, OUTPUT);
b) digitalWrite(13, OUTPUT);
c) analogRead(13);
d) pin(13, OUTPUT);
	88. What does PWM stand for?
	a) Pulse Width Modulation
	b) Power Width Method
	c) Pulse Wire Modulation
	d) Power Wave Method

89. Which bracket is used after loop in Arduino code? a) [] b) () c) {} d) <>	90. Which command turns ON an LED? a) digitalWrite(9,HIGH) b) digitalRead(9) c) analogRead(9,0) d) pinMode(0,0)



91. Which symbol is used for comments in Arduino? a) // b) ## c) %% d) @@	92. Which baud rate is commonly used? a) 9600 b) 100 c) 500 d) 120
93. Which command prints data on Serial Monitor? a) Serial.print() b) Serial.begin() c) pinMode() d) delay()	94. Which pin is generally connected to built-in LED? a) 10 b) 11 c) 12 d) 13
95. Which sensor is used in touchless sanitizer machines? a) IR Sensor b) LDR c) Relay d) Servo	96. DC motors are used in: a) Fans b) Cars c) Robots d) All of these
97. Which motor is commonly used in robotic arms? a) Servo Motor b) Fan Motor c) AC Motor d) Pump Motor	98. Which motor is used in printers? a) Servo b) Stepper c) DC d) Gear
99. Which keyword stores whole numbers? a) float b) char c) int d) string	100. Which data type stores decimal numbers? a) int b) float c) char d) boolean



Answer Key (Practice Questions):					
1→C	21→ B	31→D	51→ B	71→C	91→A
2→D	22→ C	32→B	52→ B	72→B	92→A
3→B	23→C	33→B	53→C	73→C	93→A
4→D	24→B	34→B	54→B	74→B	94→D
5→B	25→B	35→C	55→B	75→C	95→A
6→A	26→C	36→B	56→C	76→A	96→D
7→D	27→B	37→A	57→A	77→B	97→A
8→B	28→B	38→B	58→B	78→B	98→B
9→B	29→C	39→A	59→B	79→B	99→C
10→B	30→B	40→A	60→B	80→C	100→B
11→A		41→B	61→B	81→A	
12→A		42→A	62→A	82→B	
13→C		43→C	63→A	83→A	
14→D		44→D	64→C	84→C	
15→A		45→B	65→C	85→B	
16→C		46→C	66→A	86→A	
17→D		47→B	67→A	87→A	
18→B		48→C	68→A	88→A	
19→C		49→C	69→B	89→C	
20→B		40→B	60→A	60→A	



5. Model Test Paper

Tech Olympiad 2026
Group -1 (Classes 6-8)

Time: 1 Hour | Maximum Marks: 40

1. Which everyday object becomes an IoT device when it connects to the internet?
   a) A normal pencil
   b) A smart watch that sends steps to your phone
   c) A wooden chair
   d) A paper notebook

2. What do we call the parts that FEEL things like heat, light, or movement?
   a) Stickers
   b) Sensors
   c) Batteries
   d) Covers

3. A smart dustbin that says "I am full" uses IoT for which city application?
   a) Smart City - Waste Management
   b) Smart Fashion
   c) Smart Games
   d) Smart Painting

4. What is the brain of most student IoT projects that takes decisions?
   a) Breadboard
   b) Wire only
   c) Microcontroller like Arduino
   d) Glue

5. If your smart fitness band has no battery, what will happen?
   a) It will work faster
   b) It will fly
   c) It will stop sensing and sharing data
   d) It will store more data

6. Your smart home has 15 devices. Instead of connecting all 15 directly to the internet, they connect to one small box which then connects to the internet. This box is called:
   a) Main switch board for electricity
   b) Water tank
   c) Gateway - it manages and secures all devices at one place
   d) Speaker

7. A smart water tank motor keeps overflowing even though the system is IoT enabled. What is the most likely reason?
   a) Internet is too fast
   b) Cloud storage is full of photos
   c) Water level sensor is giving wrong data or is dirty
   d) Motor is too new

8. Why does a fitness band use Bluetooth Low Energy (BLE) and not Wi-Fi to talk to your phone?
   a) BLE uses very little battery, so band works for many days
   b) BLE can show YouTube videos
   c) Wi-Fi does not work with humans
   d) BLE is heavier



9. A smart light turns ON when motion sensor detects you, and turns OFF when no motion for 5 minutes. Which components are working? a) Only sensor b) Only internet cable c) Only cloud d) Motion Sensor (Input) -> Controller -> Relay Actuator (Output)	10. What does "Interoperability" problem mean for you at home? a) Your smart bulb from Company A does not work with app of Company B b) Your bulb is too bright c) Internet is too slow d) Bulb needs more wires
11. Your smart greenhouse has: Soil sensor, Temperature sensor. The rule is: Water pump ON only if Soil is DRY AND Temperature is >30°C. If soil is dry but temp is 25°C, what will system do? a) Pump will turn ON because soil is dry b) Pump will burst c) Sensors will stop working d) Pump will stay OFF because both conditions are not met	12. An IoT device first saves data when there is no internet, then sends all data when internet comes back. This is called: a) Store and Forward / Offline Sync b) Deleting data forever c) Real-time only system d) No memory system
13. Which combination makes a device truly SMART and not just AUTOMATIC? a) Both are same b) An automatic door opens on sensor. A smart door learns your timing and opens before you come c) Smart means bigger size d) Automatic means using internet	14. A student connects a big 12V water pump directly to Arduino 5V pin. What will happen? a) Arduino pin will burn because it cannot give high current and voltage - need a relay/driver module b) Pump will run perfectly and safely c) Arduino will become faster d) Pump will give Wi-Fi
15. If you send all raw data from 100 city sensors directly to cloud in USA every second, what 2 problems will occur? a) Sensors will become clean b) High data cost and High latency (delay) c) City will become colder d) No problem	16. Baud rate for Serial Communication of data is_____. a) 9500 b) 9300 c) 9600 d) 7500



17. delay(5000) function is used in Arduino code for giving_____.
    a) 50 second delay
    b) 500 second delay
    c) 5 second delay
    d) 0.5 second delay

18. What is the minimum voltage Arduino UNO Board required?
    a) 5V
    b) 9V
    c) 2V
    d) 6V

19. Which component converts electrical energy into sound?
    a) LED
    b) Buzzer
    c) Relay
    d) LDR

20. Which Arduino function is used to read the state of a push button?
    a) analogRead()
    b) digitalRead()
    c) digitalWrite()
    d) pinMode()

21. Which display is suitable for showing text and simple graphics in a compact size?
    a) Seven-Segment Display
    b) OLED Display
    c) Buzzer
    d) Relay

22. Which component limits the current flowing through an LED?
    a) Capacitor
    b) Resistor
    c) Relay
    d) Switch

23. Which sensor is commonly used in automatic hand sanitizer dispensers?
    a) Soil Moisture Sensor
    b) IR Proximity Sensor
    c) DHT11
    d) Gas Sensor

24. Which Arduino function repeats continuously after setup() finishes?
    a) start()
    b) run()
    c) loop()
    d) repeat()

25. Which of the following is an input device for Arduino?
    a) LCD Display
    b) Servo Motor
    c) Push Button
    d) Buzzer

26. Which motor can rotate to a specific angle such as 90° or 180°?
    a) DC Motor
    b) Servo Motor
    c) Gear Motor
    d) Induction Motor

27. In a smart irrigation project, when should the water pump turn ON?
    a) When the soil is sufficiently wet
    b) When the soil moisture is below the set threshold
    c) All the time
    d) Only during daytime

28. Which component stores electrical energy temporarily?
    a) Capacitor
    b) LED
    c) Resistor
    d) Switch

29. Sensors work as _____device in automation circuits.
    a) Input
    b) Output
    c) Controller
    d) programmer

30. Identify component



    a) ultrasonic sensor
    b) Gas sensor
    c) sound sensor
    d) Touch sensor

31. What is the usual size of a standard Arduino LCD display?
    a) 8 × 8
    b) 16 × 2
    c) 32 × 32
    d) 20 × 20

32. Which symbol ends every Arduino statement?
    a) .
    b) ;
    c) :
    d) ,

33. An LED connected to pin 13 is not glowing. Which should you check first?
    a) Wiring and pinMode() setting
    b) Battery colour
    c) Sensor value only
    d) Arduino cabinet

34. Which branch combines mechanics, electronics, and programming?
    a) Robotics
    b) Painting
    c) Agriculture
    d) History

35. What is the main function of a microcontroller?
    a) Play music
    b) Store images
    c) Read inputs, process data, and control outputs
    d) Increase internet speed

36. Which cable is commonly used to connect an Arduino Uno to a computer?
    a) HDMI Cable
    b) USB Cable
    c) VGA Cable
    d) Ethernet Cable



37. Which pins are generally used to connect LEDs and buttons? a) Analog Pins only b) Digital Pins c) GND Pins d) RESET Pin	38. What happens when you upload a new program to Arduino? a) Old program is replaced b) Old and new programs run together c) Arduino stops working d) Memory doubles
39. Why is Arduino popular for beginners? a) It is open-source and easy to program. b) It requires software. c) It works with robots. d) It can connect to sensors.	40. Which value represents HIGH in Arduino? a) 0 b) 1 c) -1 d) 255

Answer Key: Model Test Paper				
1→B	11→D	21→B	31→B	
2→B	12→A	22→B	32→B	
3→A	13→B	23→B	33→A	
4→C	14→A	24→C	34→A	
5→C	15→B	25→C	35→C	
6→C	16→C	26→B	36→B	
7→C	17→C	27→B	37→B	
8→A	18→A	28→A	38→A	
9→D	19→B	29→A	39→A	
10→A	20→B	30→B	40→B	